Linux Storage Management and LVM

Server Engineering Labs

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Topics Covered:

- Linux storage administration
- Partition management
- Logical Volume Management (LVM)
- Filesystem migration
- Persistent mounts with `fstab`

Environment:
Debian Linux Virtual Machine
Virtualized lab environment

Contents

1. Introduction

This report documents several Linux storage administration tasks performed in a virtualized Debian environment. The main objective was to understand how Linux handles disks, partitions, mount points, and Logical Volume Management (LVM).

The exercises focused on:

- Detecting and managing storage devices
- Creating and mounting partitions
- Configuring LVM structures
- Migrating directories such as `/var`
- Making mounts persistent through `/etc/fstab`

The work was carried out in a controlled laboratory environment using virtual machines.

2. Disk Detection and Initial Inspection

The first step consisted of inspecting the disks detected by the operating system.

Useful commands included:

```
lsblk
fdisk -l
df -h
```

These commands allow the administrator to:

- Identify physical disks
- Check existing partitions
- Verify mounted filesystems
- Inspect storage usage

Figure 1: Initial disk layout and LVM structure shown using `lsblk`.

3. Creating and Managing Partitions

A new partition was created in order to prepare additional storage for LVM.
Typical workflow:

```
sudo fdisk /dev/sdb
```

Inside `fdisk`, the partition table was modified and a new Linux partition was created.
After saving the changes, the partition table was reloaded:

```
sudo partprobe
```

The partition was then formatted:

```
sudo mkfs.ext4 /dev/sdb1
```

4. Logical Volume Management (LVM)

The laboratory also introduced the use of LVM for flexible storage administration.
The process included:

1. Creating a Physical Volume (PV)
2. Creating a Volume Group (VG)
3. Creating a Logical Volume (LV)
4. Formatting and mounting the LV

4.1. Creating the Physical Volume

```
sudo pvcreate /dev/sdb1
```

4.2. Creating the Volume Group

```
sudo vgcreate vg_data /dev/sdb1
```

4.3. Creating the Logical Volume

```
sudo lvcreate -L 5G -n lv_var vg_data
```

4.4. Formatting the Logical Volume

```
sudo mkfs.ext4 /dev/vg_data/lv_var
```

5. Migrating the `/var` Directory

One of the most important tasks was migrating the `/var` directory into the new logical volume.
The migration process required:

- Creating a temporary mount point
- Copying all files while preserving permissions
- Updating `/etc/fstab`
- Rebooting and validating the configuration

5.1. Temporary Mount

```
sudo mkdir /mnt/var_temp
sudo mount /dev/vg_data/lv_var /mnt/var_temp
```

5.2. Copying Files

```
sudo rsync -av /var/ /mnt/var_temp/
```

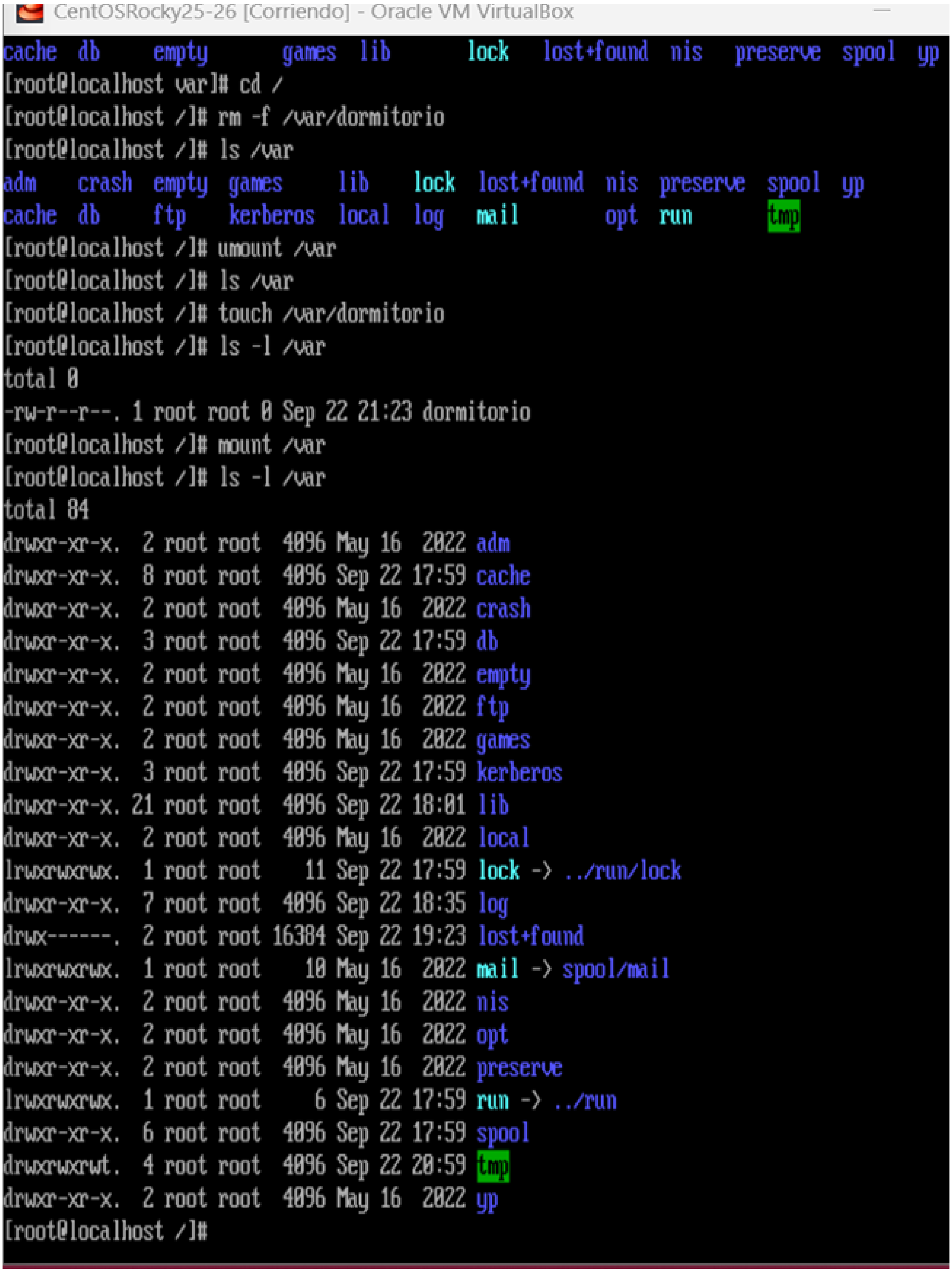

Figure 2: Validation of the temporary `/var` mount and filesystem behavior after unmounting and remounting operations.

5.3. Persistent Mount Configuration

The filesystem was added to `/etc/fstab`:

```
UUID=<uuid> /var ext4 defaults 0 2
```

6. Verification and Validation

After rebooting the system, the configuration was verified using:

```
mount | grep var
df -h
lsblk
```

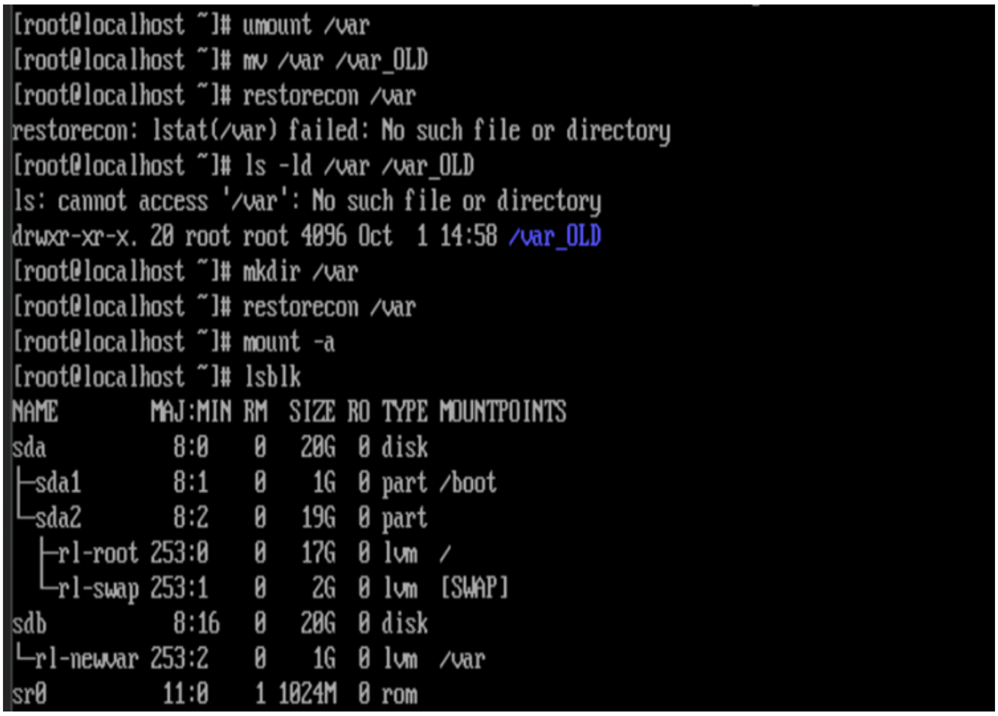

Figure 3: Final LVM configuration after mounting the logical volume on `/var`.

The logical volume was successfully mounted and persisted correctly after reboot.

7. Challenges Encountered

Several issues appeared during the lab:

- Incorrect mount configurations
- Filesystem permission inconsistencies
- Risks when modifying `/etc/fstab`

Particular care had to be taken because incorrect `fstab` entries can prevent Linux from booting correctly.

8. Conclusion

This laboratory provided practical experience with Linux storage administration and LVM management.

The exercises demonstrated:

- How Linux organizes storage devices
- How logical volumes improve flexibility

- How to safely migrate important directories
- The importance of persistent filesystem configuration

These concepts are essential for Linux system administration, server deployment, and infrastructure management.